A Geometric Interpretation of Liouville's Formula

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The solutions of a linear differential system define a family of linear transformations of the space of initial conditions. This note explains how the determinant of the fundamental matrix measures the resulting change in volume and why its rate of change is governed by the trace of the coefficient matrix. An elementary derivation of Liouville's formula is followed by a geometric interpretation and two planar examples. The prerequisites are basic linear algebra, multivariable calculus, and first-order differential equations.

1 Introduction

A differential equation describes the motion of an individual point once an initial condition has been chosen. It also describes the motion of an entire region when all its points are allowed to evolve according to the same equation. For a linear system, this motion is determined by a matrix, so the geometric effect can be studied using determinants. The question addressed here is how the volume of a region changes as it moves under the system.

The relevant result is usually called *Liouville's formula*, and is also known as Abel's identity in the theory of linear differential equations [1, Sec. 3.4]. It relates the determinant of a fundamental matrix to the trace of the matrix defining the system. The derivation below follows the geometric route: first describe the evolving region, then apply the change-of-variables formula, and finally differentiate the determinant using Jacobi's formula [2, p. 1]. All arguments remain within the setting of real linear systems.

2 Linear systems and evolving regions

We begin by relating the solutions of a linear system to a transformation of its initial conditions. This requires fixing the initial time and choosing the normalization of the fundamental matrix explicitly.

Let $I \subseteq \mathbb{R}$ be an interval, let $t_0 \in I$, and let $A(t)$ be a real $n \times n$ matrix whose entries are continuous on I. The state $\boldsymbol{x}(t) \in \mathbb{R}^n$ satisfies the initial-value problem

$$\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = A(t)\boldsymbol{x}(t), \qquad \boldsymbol{x}(t_0) = \boldsymbol{x}_0. \tag{2.1}$$

The existence and uniqueness theorem for linear systems guarantees a unique solution on I for each initial vector $\boldsymbol{x}_0$ [1, Sec. 3.4]. A fundamental matrix has linearly independent solution vectors as its columns. Writing I_n for the identity matrix, we choose the normalized fundamental matrix $U(t)$ defined by

$$U'(t) = A(t)U(t), \qquad U(t_0) = I_n, \tag{2.2}$$

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Its columns are the solutions starting from the standard basis vectors. Linearity then gives the solution with arbitrary initial data as

$$\boldsymbol{x}(t) = U(t)\boldsymbol{x}_0. \tag{2.3}$$

The condition $U(t_0) = I_n$ is what makes $\boldsymbol{x}_0$ the actual initial position in Eq. (2.3). For an unnormalized fundamental matrix $Y(t)$, the corresponding expression would be $\boldsymbol{x}(t) = Y(t)Y(t_0)^{-1}\boldsymbol{x}_0$.

The matrix $U(t)$ is invertible at every time in I. Indeed, if $U(t_*)\boldsymbol{c} = \boldsymbol{0}$ for some $t_* \in I$, the solution $U(t)\boldsymbol{c}$ passes through the origin at t_*. Uniqueness implies that this is the zero solution throughout I, and evaluation at t_0 gives $\boldsymbol{c} = \boldsymbol{0}$. Thus $U(t)$ has a trivial kernel. Its determinant is continuous and never vanishes; since $\det U(t_0) = 1$, it is positive throughout I.

At a fixed time, the velocity field is $\boldsymbol{v}(t, \boldsymbol{x}) = A(t)\boldsymbol{x}$. Along a solution, its value is $A(t)U(t)\boldsymbol{x}_0$, which equals the tangent vector $\boldsymbol{x}'(t)$. If A is constant, this field is independent of time. The map in Eq. (2.3) is a linear deformation of the space of initial conditions. It fixes the origin and can stretch, compress, rotate, or shear a region.

Let $\Omega_0 \subset \mathbb{R}^n$ be a bounded region with a piecewise smooth boundary and positive volume. Its image at time t and its volume are denoted by

$$\Omega_t = U(t)\Omega_0, \qquad V(t) = \int_{\Omega_t} \mathrm{d}^n\boldsymbol{x}, \qquad V_0 = V(t_0), \tag{2.4}$$

where $\mathrm{d}^n\boldsymbol{x} = \mathrm{d}x_1 \cdots \mathrm{d}x_n$ is the usual volume element. For $n = 2$, the word volume refers to area. The Jacobian matrix of the map $\boldsymbol{x}_0 \mapsto U(t)\boldsymbol{x}_0$ is $U(t)$ itself. Hence the change-of-variables formula gives

$$\begin{aligned} V(t) &= \int_{\Omega_0} |\det U(t)| \, \mathrm{d}^n\boldsymbol{x}_0 \\ &= \det U(t)\, V_0. \end{aligned} \tag{2.5}$$

The absolute value is required in the change-of-variables formula. It can be removed here because the normalized fundamental matrix has positive determinant. Since $U(t)$ does not depend on the initial point, its determinant is the same volume factor everywhere in the region.

3 Liouville's formula

Equation (2.5) reduces the volume problem to a scalar quantity. We write $\varphi(t) = \det U(t)$ and derive a differential equation for φ. The matrix operation that appears in this equation is the trace. For an $n \times n$ matrix $B = (b_{ij})$, it is defined by

$$\operatorname{tr} B = \sum_{i=1}^{n} b_{ii}. \tag{3.1}$$

For two $n \times n$ matrices B and C, summing their diagonal entries shows that $\operatorname{tr}(BC) = \sum_{i,j} b_{ij}c_{ji} = \operatorname{tr}(CB)$. This is the cyclic property of the trace; it does not assert that $BC = CB$.

To differentiate φ, let $C_{ij}(t)$ denote the cofactor of the entry $U_{ij}(t)$. Cofactor expansion gives $\partial(\det U)/\partial U_{ij} = C_{ij}$, while the inverse-matrix formula gives $C_{ij} = \det U\,(U^{-1})_{ji}$. Applying the chain rule therefore yields

$$\begin{aligned} \varphi'(t) &= \sum_{i,j=1}^{n} C_{ij}(t)U'_{ij}(t) \\ &= \varphi(t) \sum_{i,j=1}^{n} (U(t)^{-1})_{ji}U'_{ij}(t) \\ &= \varphi(t)\, \operatorname{tr}\Big(U(t)^{-1}U'(t)\Big). \end{aligned} \tag{3.2}$$

This identity is Jacobi's formula [2, p. 1]. The preceding invertibility argument justifies the use of $U(t)^{-1}$.

Substituting $U' = AU$ from Eq. (2.2) makes the differential equation for the determinant explicit. The cyclic property gives $\operatorname{tr}(U^{-1}AU) = \operatorname{tr}(AUU^{-1}) = \operatorname{tr} A$, so Eq. (3.2) becomes

$$\varphi'(t) = \operatorname{tr} A(t)\, \varphi(t), \qquad \varphi(t_0) = 1. \tag{3.3}$$

Because $\varphi(t) > 0$, we may divide by it and integrate. This gives

$$\log \varphi(t) - \log \varphi(t_0) = \int_{t_0}^{t} \operatorname{tr} A(s)\, \mathrm{d}s. \tag{3.4}$$

Exponentiating and recalling Eq. (2.5) completes the derivation. We state the result with both its matrix and geometric forms.

Theorem 3.1 (Liouville's formula)**.** *Let $A : I \to \mathbb{R}^{n\times n}$ be continuous, and let U satisfy Eq. (2.2). Then, for every $t \in I$,*

$$\det U(t) = \exp\left(\int_{t_0}^{t} \operatorname{tr} A(s)\, \mathrm{d}s \right). \tag{3.5}$$

For the evolving region in Eq. (2.4), the volume consequently satisfies

$$V'(t) = \operatorname{tr} A(t)\, V(t), \tag{3.6a}$$

$$V(t) = V_0 \exp\left(\int_{t_0}^{t} \operatorname{tr} A(s)\, \mathrm{d}s \right). \tag{3.6b}$$

Thus the volume can be determined from the trace without finding every entry of the fundamental matrix. For a constant matrix A, the integral reduces to $(t - t_0) \operatorname{tr} A$. The same derivation covers continuous time-dependent matrices.

4 Geometric meaning of the trace

The trace has a direct interpretation as the instantaneous relative rate of volume change. This follows by dividing Eq. (3.6a) by the positive quantity $V(t)$. A positive trace gives increasing volume at that time, and a negative trace gives decreasing volume. If $\operatorname{tr} A(t) = 0$ for every $t \in I$, every evolving region preserves its volume. Conversely, if a region of positive volume has constant volume throughout I, Eq. (3.6a) implies that the trace vanishes throughout I.

The same quantity appears in elementary vector calculus. If $A(t) = (a_{ij}(t))$, the components of the velocity field are $v_i(t, \boldsymbol{x}) = \sum_{j=1}^{n} a_{ij}(t) x_j$. Its divergence, defined as the sum of the derivatives of each component with respect to the matching coordinate, is

$$\nabla_{\boldsymbol{x}} \cdot \boldsymbol{v}(t, \boldsymbol{x}) = \sum_{i=1}^{n} \frac{\partial v_i}{\partial x_i} = \sum_{i=1}^{n} a_{ii}(t) = \operatorname{tr} A(t). \tag{4.1}$$

For this linear field, the divergence is spatially uniform, so the same relative volume change occurs throughout the region. A zero trace at a single instant only makes the volume derivative vanish at that instant; volume preservation over an interval requires it to vanish throughout the interval. Volume preservation allows lengths and angles to change, as the following examples demonstrate.

5 Two elementary examples

Planar systems make the determinant visible as an area factor. We set $t_0 = 0$ and use the unit square $\Omega_0 = [0, 1]^2$ as the initial region. The first example allows independent changes along the coordinate axes; the second gives a shear. Both can be solved by direct integration.

5.1 Expansion and compression

Let a and b be real constants and consider the equations $x' = ax$ and $y' = by$. The coefficient matrix and the normalized fundamental matrix are

$$A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, \qquad U(t) = \begin{pmatrix} e^{at} & 0 \\ 0 & e^{bt} \end{pmatrix}. \tag{5.1}$$

The unit square becomes a rectangle with side lengths e^{at} and e^{bt}. Its area is therefore

$$V(t) = e^{(a+b)t}, \qquad \operatorname{tr} A = a + b. \tag{5.2}$$

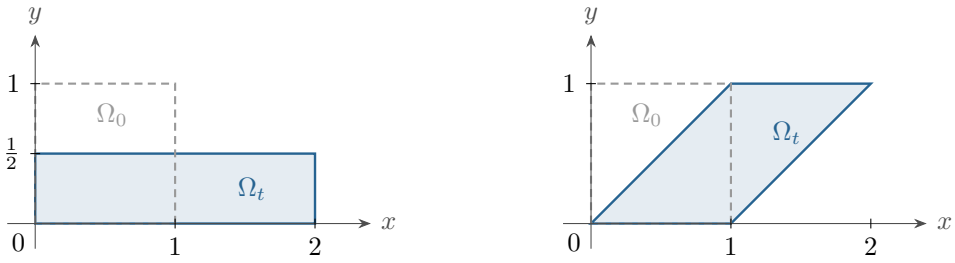
This agrees with Eq. (3.6b). The area increases when $a + b > 0$, decreases when $a + b < 0$, and remains constant when $a + b = 0$. For the choice $a = 1$ and $b = -1$, expansion along the horizontal direction exactly compensates compression along the vertical direction. At $t = \log 2$, the sides have lengths 2 and $1/2$, so the area remains equal to one. Figure 1(a) shows this deformation.

5.2 Shear with constant area

Let k be a real constant and consider $x' = ky$ and $y' = 0$. Since the vertical coordinate stays fixed, direct integration gives $y(t) = y_0$ and $x(t) = x_0 + kty_0$. In matrix form,

$$A = \begin{pmatrix} 0 & k \\ 0 & 0 \end{pmatrix}, \qquad U(t) = \begin{pmatrix} 1 & kt \\ 0 & 1 \end{pmatrix}. \tag{5.3}$$

Here $\operatorname{tr} A = 0$ and $\det U(t) = 1$, so the area is constant. The square becomes a parallelogram whose vertices are $(0, 0)$, $(1, 0)$, $(1 + kt, 1)$, and $(kt, 1)$. Its base and height both have length one, although its sloping sides change length. Figure 1(b) shows the case $kt = 1$.



(a) Expansion and compression **(b)** Shear

Figure 1: Two deformations of the unit square. The dashed outline is the initial region Ω_0 and the shaded region is its image Ω_t. In (a), $U = \operatorname{diag}(2, 1/2)$; in (b), $U = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Both matrices have determinant one, so both images have unit area. The coordinate scales are identical in the two panels.

These examples illustrate the two roles of the matrices in the argument. The fundamental matrix $U(t)$ determines the full deformation, and its determinant gives the resulting volume factor. The coefficient matrix $A(t)$ determines the instantaneous velocity field, and its trace gives the relative rate at which that volume changes.

References

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